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## **Enhancing Risk Based Inspection Strategies for Safe Smart Ex Operations**

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### **Abstract**

Traditional hazardous area (Ex) inspection models are increasingly unfit for modern offshore operations. They are hindered by paper-based processes, rigid scheduling, and siloed execution. This paper introduces the EX Roadmap, a structured three-stage approach to achieving Risk-Based Inspection (RBI) maturity. The model combines digital foundations, engineering intelligence, and continuous supervision to deliver safer, more efficient, and sustainable inspection practices. The findings demonstrate up to a 48% reduction in inspection effort, enhanced regulatory compliance, and measurable improvements in asset integrity.

### **Introduction: From Compliance to Intelligence**

In 2011, the organisation considered itself an expert in Ex compliance. Inspection regimes were thorough, aligned with international standards, and consistently applied across every Ex-certified asset in the fleet. On paper, every box was ticked, and there was a firm belief that strict adherence to the rules meant people and infrastructure were well protected.

Then came a pivotal moment.

There was no explosion and no catastrophic failure—just one subtle, easily overlooked incident that quietly revealed a blind spot in an otherwise diligent system. That moment highlighted an uncomfortable truth: compliance does not guarantee protection. Checking boxes is not the same as managing risk. No matter how comprehensive, a checklist cannot replace the dynamic insight required to understand risk in complex, high-stakes environments.

This incident sparked a fundamental shift in approach. A thorough internal investigation was launched, drawing on insights from operations, engineering, safety, and maintenance teams. The conclusion was unmistakable: the risk had not arisen from negligence, but from a lack of context. Defects were being treated the same—regardless of where the asset was located, how it was used, or what consequences a failure might have.

The system relied on static, paper-based records and fragmented datasets. Inspections were conducted on fixed schedules, disconnected from actual operational conditions. What was missing was the intelligence needed to differentiate critical threats from routine issues before they escalated.

That realisation marked the beginning of a transformation—from procedural compliance to operational intelligence, where inspections became a proactive defence layer powered by data and driven by purpose.

## Why This Matters for the Future

In hazardous environments such as Floating Production Storage and Offloading (FPSO) units, even minor oversights can have major consequences. The evolved approach is designed not just to meet minimum standards but to anticipate, adapt, and intervene—before failures occur.

Resilience is being built by:

- Empowering frontline teams with real-time, contextualised data
- Leveraging automation and analytics to reduce variability and human error
- Embedding a learning culture that values insight over box-ticking
- Aligning inspections to actual, evolving risk—not rigid calendar schedules

This represents far more than a tactical upgrade—it is a strategic reinvention that places people, data, and intelligent systems at the core of operational excellence.

## In Summary

This report traces the journey—from a quiet failure to a future-focused model that positions inspection as a critical enabler of safety, reliability, and trust. It offers a blueprint for organisations aiming to move beyond regulatory minimums and adopt a smarter, more resilient approach to asset integrity.

The vision represents the future of Ex management: a collaborative ecosystem where technology enhances human judgment, where data is transformed into informed decisions, and where protection is deliberately proactive rather than reactive.

This is not merely a retrospective—it is a forward-looking framework for those who recognise that genuine safety, reliability, and performance come from understanding risk in its true context.

The transition has been made from procedural compliance to intelligent, risk-based inspection; from paper records to integrated digital platforms; from isolated insights to connected, enterprise-wide intelligence.

This is the new benchmark for Ex integrity—where people are empowered by data, systems are optimised according to consequence, and risks are managed before they escalate.

It is time to stop chasing compliance and start designing for resilience.

## Vision and Strategy: Shaping a Data-Driven Future

### From Compliance to Continuous Improvement

The organisation's strategic transformation began with a shift in mindset. Inspection was reimagined not as a compliance obligation but as a strategic opportunity—one capable of unlocking operational value, reducing downtime, and strengthening safety.

This was not a passive evolution; it required bold investment in three critical areas:

- **People** – Structured training and active involvement in shaping inspection strategy empowered teams to take ownership of change.
- **Processes** – Workflows were redesigned to prioritise learning, adaptability, and efficiency.
- **Technology** – Digital tools were introduced to integrate insights, automate logic, and drive smarter actions.

Cross-functional collaboration played a pivotal role. Operations, engineering, safety, and maintenance functions no longer worked in silos but became co-authors of a unified approach.

The outcome was a scalable, risk-based inspection (RBI) framework that has since matured into a vision for the continuous supervision of Ex assets.

### The Power of Data

At the heart of this transformation was a resource already in place but not yet fully harnessed: data.

By systematically gathering, integrating, and interpreting operational data, the organisation was able to:

- Understand real-world context, beyond equipment specifications
- Prioritise inspection and remediation activities based on actual risk and consequence
- Eliminate unnecessary, low-value interventions
- Predict and prevent failures rather than simply react to them

This intelligence became the foundation of the EX Roadmap—a phased transformation strategy tested and proven across the global fleet. Its deployment demonstrated that intelligent inspection is not only possible but essential for any organisation seeking to modernise and de-risk operations.

### Understanding Ex: The Real Risk

Ex (explosion-protected) equipment is specifically engineered to eliminate potential ignition sources within hazardous areas—environments where flammable gases, vapours, or dust are present at concentrations capable of triggering explosions.

Such environments are commonplace in industries including oil and gas, chemicals, and offshore energy. They are particularly prevalent on Floating Production Storage and Offloading (FPSO) units, where limited space and high operational complexity significantly increase the challenge.

In these conditions, even seemingly minor defects—such as a loose gland, a missing label, or a corroded junction box—can escalate into major threats. A single uncontrolled spark or arc may result in catastrophic consequences.

Ex safety is therefore not optional; it is fundamental. Crucially, context determines risk. The same fault in two different locations can carry vastly different levels of consequence.

For this reason, inspections cannot remain purely routine. They must be intelligent, contextual, and firmly risk-informed.

### The Limits of Traditional Inspection Models

Traditional inspection models, although developed with sound intentions, are increasingly unsuited to the demands of modern operations. Their key limitations include:

- **Paper-based reporting** – Restricts traceability, slows response times, and limits actionable insight
- **Fixed schedules** – Fail to account for operational realities such as asset condition, usage, and location
- **Subjective assessment** – Lacks the benefit of digital decision support and data-driven triangulation
- **Siloed execution** – Operates in isolation from wider maintenance, risk, and planning systems

This outdated approach creates a paradox: significant resources are spent on low-risk checks, while genuine threats often remain undetected.

The critical challenge is clear: even the smallest overlooked detail could result in injury, an unplanned shutdown, or the loss of a licence to operate.

This realisation provided the foundation for the development of a new, intelligence-led inspection strategy.

## Strategic Shift: Smarter, Safer, Future-Ready

The transformation is underpinned by ten strategic pillars designed to deliver long-term, enterprise-wide value:

1. Strengthening integrity and reliability across all Ex assets
2. Reducing operational risk and improving safety outcomes
3. Implementing risk-based inspection intervals to eliminate inefficiencies
4. Building cross-role Ex awareness and technical competency
5. Converting operational data into actionable insight
6. Digitalising inspection, reporting, and repair processes
7. Sharing lessons learned across the fleet to accelerate improvement
8. Validating solutions internally before external deployment
9. Optimising field resources and reducing site burden
10. Advancing sustainability by avoiding unnecessary interventions and emissions

## Why This Matters for the Future

On FPSOs and similar high-consequence environments, a reactive approach is no longer acceptable. The RBI-led model ensures:

- Faster, smarter decision-making through contextualised data
- Greater reliability via early intervention and prioritised action
- Improved compliance and increased regulator confidence
- Enhanced asset longevity and optimised lifecycle costs

This marks a decisive shift away from static procedures towards a system that thinks, learns, and adapts—mirroring the dynamic operations it supports.

## Timeline of Transformation

The RBI strategy has been developed iteratively, refined through trial, learning, and implementation:

- **2012** – Digital inspection system introduced
- **2013** – Manual criticality ranking adopted
- **2014/15** – Standardisation of inspections with dropdowns
- **2016** – Custom RBI configuration launched
- **2017** – Field-testing and refinement of RBI logic
- **2018** – Mobile RFID/NFC tools introduced
- **2019** – Proof-of-concept for Non-Intrusive Inspection (NII)
- **2020–2022** – Integration of mechanical Ex and continuous improvements

## Understanding Risk-Based Inspection (RBI) and Its Importance

### What is Risk-Based Inspection (RBI)?

Risk-Based Inspection (RBI) is a strategic, intelligence-led approach to asset integrity and safety management. Unlike traditional inspection programmes, which apply the same fixed intervals to all equipment, RBI directs inspection and maintenance efforts where they are most needed: on assets with the highest risk of failure or consequence.

In hazardous and complex environments—such as offshore oil and gas facilities, refineries, and chemical plants—RBI shifts the focus from calendar-driven checks to targeted, risk-informed interventions. It addresses three critical questions:

- Which equipment is most likely to fail?
- What would be the consequence of that failure?
- Where should inspection resources be directed to reduce the greatest risk?

This is particularly vital in hazardous (Ex) zones, where explosion-protected equipment must prevent ignition sources without fail. An overlooked loose connection, degraded seal, or damaged cable could escalate into a catastrophic incident. RBI ensures inspection resources are directed to these potential ignition points based on actual risk, rather than arbitrary schedules.

### Why RBI Matters

Traditional inspection programmes rely on fixed schedules that ignore asset condition, operating environment, or criticality. While they may satisfy minimum compliance requirements, they do not effectively manage evolving risk.

RBI provides a stronger alternative by:

- Identifying and addressing critical risks early, preventing escalation
- Improving efficiency by targeting high-risk equipment and reducing unnecessary checks
- Extending asset life through timely, risk-based maintenance
- Demonstrating proactive risk management to regulators and stakeholders
- Minimising operational disruption by reducing offshore personnel exposure, intrusive inspections, and associated carbon footprint

In essence, RBI transforms inspection from a reactive, compliance-driven activity into a proactive, data-informed strategy that safeguards people, protects production, and adds measurable value throughout the asset lifecycle.

### Why RBI is Essential in Hazardous Areas

On offshore production facilities and other hazardous sites, the stakes are exceptionally high. Downtime is costly, but the cost of safety failure is immeasurably greater. Ex zones, where flammable gases or vapours may be present, demand the strictest controls to prevent ignition.

Traditional approaches often create two problems:

- **Over-inspection** – Excessive checks on low-risk assets waste time and resources without improving safety
- **Under-inspection** – Critical equipment in high-risk zones may be neglected for too long, increasing failure likelihood

RBI resolves these issues by considering multiple factors—location, environmental exposure, fault history, equipment type, and operational criticality—to prioritise assets accurately. The result is a focused, data-informed strategy aligned with actual operational risk.

### **Breaking Away from Outdated Practices**

Despite advances, more than half of offshore Ex inspections are still carried out with pen and paper—susceptible to inconsistency, error, and poor traceability. Fixed schedules and analogue methods may meet legal requirements, but they provide only the illusion of safety.

The central challenge became clear:

**"How can inspections move beyond box-ticking compliance to become a strategic driver of safety, integrity, and operational excellence?"**

RBI provides the answer. It replaces reactive, outdated routines with a forward-looking, intelligent model rooted in live operational risk. Inspectors, engineers, and decision-makers are supported by digital tools and robust data, enabling safer, faster, and more consistent outcomes.

### **How RBI Works: Five Key Steps**

#### **1. Criticality Database**

Every asset (valves, sensors, panels, motors, etc.) is recorded in a secure system, assigned inspection requirements and a criticality score based on failure consequences.

#### **2. Inspection Results**

Inspectors use tablets or mobile devices to scan RFID tags on equipment, logging Pass/Fail outcomes directly into the system.

#### **3. Risk Calculation**

Where failures are detected, the system calculates a repair priority score by evaluating:

- Component criticality
- Ignition risk (voltage or spark energy potential)
- Hazardous zone classification
- Primary and secondary location risks

Higher scores equate to higher repair priority.

#### **4. Repair and Maintenance Planning**

A repair priority list guides maintenance teams to address the most critical issues first, reducing uncertainty and improving safety.

#### **5. Smarter Scheduling**

Inspection intervals are set dynamically:

- High-risk items may require checks every 6–12 months
- Low-risk items may be safely extended to 4–5 years

This tailored model extends asset life, enhances compliance, and minimises unplanned downtime.

## **The EX Roadmap: Enabling RBI Success through a Structured Three-Stage Model**

Achieving Risk-Based Inspection (RBI) success is not a one-time shift but a progressive journey. The EX Roadmap provides a structured, future-proofed pathway for transforming hazardous area (Ex) inspection



practices from fragmented, compliance-driven routines into a smart, sustainable model underpinned by data, digitalisation, and engineering intelligence.

This transformation is delivered through three distinct but interconnected stages, referred to as Pack 1, Pack 2, and Pack 3. Each Pack builds upon the last, generating momentum towards safer operations, cost optimisation, and continuous regulatory confidence. While each stage can be deployed independently, the full value of the roadmap is realised when executed sequentially.

By applying all three Packs, organisations are able to unlock the full potential of Risk-Based Inspection — achieving measurable improvements in safety, productivity, and asset integrity.

### **Pack 1: Digital Foundation – Pre-Survey & Baseline Creation**

**Purpose.** To establish the digital infrastructure for Ex compliance through comprehensive equipment tagging, data capture, and the creation of an intelligent Hazardous Area Equipment Register (HAER). This Pack provides the groundwork for future RBI implementation.

**Summary.** Pack 1 initiates the shift from manual, reactive inspection regimes to digital asset intelligence. By building a centralised, verified HAER and embedding RFID tagging, it creates an accurate, searchable, and inspection-ready database — the essential first step for applying RBI logic.

#### ***Key Deliverables.***

- RFID tagging of Ex equipment for traceability and efficiency
- Digital HAER with zone classification, protection type, and operational metadata
- Baseline inspection (sweep survey) to confirm condition and validate records
- Risk categorisation for RBI readiness (low, medium, high)
- Digital platform setup including user testing, pilot programme, and field tools

#### ***Investment Areas.***

- One-time field effort for inspection, tagging, and digitisation
- Procurement or integration of Ex-compliant inspection software
- Technician onboarding and system training

#### ***People Outcomes.***

- Faster technician workflows enabled by digital tools
- Reduced burden on Ex Responsible Persons (ExRPs)
- Remote visibility of equipment condition and risk

#### ***Strategic Benefits.***

- Up to 45% reduction in site inspection effort
- High accuracy and audit-readiness of HAER
- Digital backbone for RBI implementation
- Improved OPEX targeting from the outset

## **Pack 2: Engineering Intelligence – RBI Enablement & Data-Driven Integrity**

**Purpose.** To embed risk-based thinking and engineering logic into inspection planning, optimising frequencies, reducing unnecessary inspections, and prioritising safety-critical equipment.

**Summary.** Pack 2 accelerates transformation by converting data into actionable intelligence — enabling targeted inspections, smarter repairs, and a step-change in inspection efficiency. Risk scores are dynamically calculated, and inspection tasks are aligned with actual operational threats rather than fixed calendar cycles.

### ***Key Deliverables.***

- Strategic oversight from PMT to guide rollout and execution
- Data enrichment to fill gaps and validate fields within HAER
- RBI scoring using algorithms that assess fault history, zone exposure, and criticality
- Smart repair and reinspection logic that adapts to risk, not time
- Modular work-pack scheduling that groups tasks geographically and by priority
- Training and change management to build internal RBI awareness and capability

### ***Investment Areas.***

- Engineering effort to review and enhance data
- Upskilling technicians and engineers in RBI methodology
- Adjustment in planning and documentation practices

### ***People Outcomes.***

- ExRPs and planners concentrate on high-risk, high-value areas
- Engineers gain confidence in applying RBI logic in field decisions
- Stronger collaboration between asset integrity and operations teams

### ***Strategic Benefits.***

- 40–48% reduction in site-based inspection activity
- Up to 30% reduction in repeat inspections of low-risk assets
- RBI cycles extended up to 5 years for low-consequence equipment
- Stronger audit responses and enhanced traceability
- Clear visibility of assets ready for continuous supervision (Pack 3)

Pack 2 becomes the operational engine room of RBI, transforming baseline data into a dynamic system of risk-led inspection intelligence.

## **Pack 3: Continuous Supervision – Advanced NII & Self-Sustaining Compliance**

**Purpose.** To establish a self-sustaining model of ongoing Ex compliance through advanced non-intrusive inspection (NII) methods, skilled supervision, and intelligent sampling — reducing effort while improving outcomes.



**Summary.** Pack 3 represents the highest level of maturity within the EX Roadmap. It enables organisations to move towards proactive, real-time supervision — reducing site effort while maintaining continuous oversight. Inspection plans adapt dynamically, sampling techniques replace intrusive surveys, and skilled internal teams sustain compliance without external dependency.

***Key Deliverables.***

- NII integration for sample-based inspection with no shutdowns
- Continuous supervision managed by trained in-house personnel
- Annual inspection plan adjustments based on live data and findings
- Competency frameworks embedded across technical roles
- Digital reporting and PMR capture with real-time dashboards and regulatory traceability

***Investment Areas.***

- Development of in-house Ex supervisory capability
- Adoption of NII tools and sampling protocols
- Refinement of inspection strategy and audit readiness

***People Outcomes.***

- Lighter workloads focused on high-value tasks
- Greater confidence and autonomy across technical teams
- Reduced reliance on external vendors or specialist support

***Strategic Benefits.***

- Reduction in on-site inspection workload to 20%
- Major cost savings and reduced personnel-on-board (POB) requirements
- Seamless compliance with minimal operational disruption
- Sustained audit-readiness and regulator confidence
- Fully embedded safety culture across the workforce

Pack 3 unlocks a self-managed, data-driven Ex inspection model — one in which intelligence, not repetition, drives performance.

**RBI Success Enabled Through the EX Roadmap**

When implemented sequentially, Pack 1, Pack 2, and Pack 3 transform inspection from a paper-based, calendar-driven process into a smart, scalable, risk-led system

| Pack | Focus Area                       | People Investment                      | Technical Output                                      | Site Effort Reduction | Strategic Benefit                           |
|------|----------------------------------|----------------------------------------|-------------------------------------------------------|-----------------------|---------------------------------------------|
| 1    | Digitalisation & Baseline Survey | RFID tagging, system training          | HAER, risk categorisation, baseline inspections       | 45%                   | Visibility, audit-readiness, risk profiling |
| 2    | RBI & Data Intelligence          | Engineering upskilling, planning shift | Dynamic scheduling, risk-based prioritisation         | 40–48%                | Smarter inspection cycles, targeted repairs |
| 3    | Advanced NII & Supervision       | Competency development                 | Real-time supervision, sampling logic, PMR dashboards | 80%                   | Sustainable compliance, minimal disruption  |

The Result: A fully matured, risk-based Ex inspection system that delivers enhanced safety, cost savings, and regulatory confidence — from Day One to Year Five and beyond.

## The Benefits of a Risk-Based Approach to Ex Inspections

### 1. Contextualised Fault Management

Under a risk-based approach, faults are no longer treated equally. Each issue is assessed in terms of its location, potential consequences, and operational impact. This enables improved prioritisation and tighter risk control.

### 2. Targeted Repairs with Measurable Impact

High-consequence defects are escalated for immediate intervention, while lower-risk issues are logged, monitored, and scheduled appropriately. This ensures that safety is maintained without placing unnecessary burden on maintenance teams.

### 3. Smarter, Risk-Based Scheduling

Inspection intervals are determined by dynamic risk scoring rather than fixed calendar cycles. As a result, inspection effort is directed towards the assets that present the highest risk, rather than those that have simply reached a time threshold.

### 4. Enhanced Mobile Inspection Tools

Inspectors are equipped with intrinsically safe tablets and mobile applications to scan RFID tags, access procedures, and record findings in real time. This eliminates paperwork, streamlines workflow, and ensures that captured data is immediately actionable.

### 5. Reduced Human Error Through Decision Support

Digital checklists, logic-based prompts, and embedded inspection guidance ensure consistency in inspections. RBI tools support inspectors in making decisions aligned with organisational risk policy, reducing variability and error.

### 6. Predictive Maintenance and Trend Analysis

Live and historical data are analysed to identify early signs of asset degradation. This predictive capability enables faults to be anticipated and prevented before they lead to downtime or incident.

### 7. Long-Term Asset Integrity

Rather than focusing on short-term fixes, RBI promotes lifecycle thinking. Recurring failures are monitored, root causes identified, and long-term solutions embedded into inspection strategy, strengthening overall equipment resilience.

### 8. Enhanced Safety Outcomes

By identifying high-risk ignition sources before failure, RBI significantly reduces the likelihood of fire or explosion. In hazardous environments, this proactive risk control is vital for protecting life.

## 9. Operational Efficiency and Focus

Unnecessary repetition is eliminated, planning becomes more precise, and manpower is concentrated on the areas of highest consequence and value. The result is a leaner and more intelligent inspection regime.

## 10. Best-Practice Compliance

RBI aligns with international standards such as IEC 60079-17 and ISO/IEC 80079. This not only strengthens compliance but also provides regulators with greater confidence in organisational systems.

## 11. Campaign Optimisation and Cost Control

Smaller, more targeted inspection teams complete work more quickly and with fewer mobilisations, reducing campaign cost as well as operational disruption.

## 12. Lower Emissions and Reduced Offshore Travel

By removing unnecessary offshore visits, RBI contributes directly to decarbonisation. Campaigns are streamlined, logistics simplified, and POB (Personnel on Board) better managed—supporting ESG commitments.

## The Outcome: What RBI Delivers

- **Improved Safety:** Risk-led inspections identify and address high-consequence threats before escalation.
- **Enhanced Efficiency:** Teams achieve more with less effort, focusing on meaningful results.
- **Stronger Compliance:** Aligned with industry regulations, supported by traceable, audit-ready data.
- **Operational Confidence:** Evidence-based decisions that are consistent and defensible.
- **Lower Total Cost:** Reduced rework, optimised manpower, and fewer inspections lower overall asset cost.
- **Digital Integration:** Live data flows seamlessly between systems, enabling proactive maintenance and real-time oversight.
- **Proactive Culture:** Decisions are driven by probability and consequence rather than routine or habit.

## RBI in Action: Real-World Example – FPSO Operations

Onboard a Floating Production Storage and Offloading (FPSO) vessel, environmental stressors such as saltwater spray, vibration, UV exposure, and extreme temperatures place electrical and mechanical assets under continuous strain. Traditional inspection routines often fail to capture this nuance.

For example:

- A **lighting fixture** in a Zone 1 area near the flare vent, with exposed cable glands, elevated voltage, and a history of corrosion, is identified by RBI as high-risk and prioritised for early intervention.
- A **sensor** located inside a climate-controlled cabinet, operating at low voltage, is categorised as low-risk, requiring less frequent inspection.

This dynamic, context-driven prioritisation is what sets RBI apart—it directs time and budget where it truly matters.

## Understanding Context: One Fault, Two Outcomes

A missing cable label might be found during an inspection.

- **Scenario A: Laydown Area** – open, ventilated, low likelihood of gas → Low repair priority (120-day fix).

- **Scenario B: Pump Room** – enclosed space, high likelihood of gas, critical process equipment nearby → Urgent repair (60-day fix or immediate mitigation).

Without RBI, both faults may have been treated equally. With RBI, context dictates action, ensuring safety and operational performance are not compromised.

### **RBI in Practice: Real Inspection Scenario**

A technician inspects a damaged valve in the pump room:

- High-voltage
- Located in a Zone 1 hazardous area
- Surrounded by other live systems

RBI logic escalates this for immediate repair. By contrast, a similar valve on deck with lower risk exposure would be scheduled for later intervention.

### **Smart Scheduling with RBI**

- Critical assets may require 6–12 month cycles.
- Low-risk assets may extend to 3–5 year cycles.

This shift from routine to rationale ensures resources are allocated exactly where they are most needed.

### **Strategic Shift: From Subscription to Performance-Based Services**

The RBI model, strengthened by digital transformation, supports a shift away from traditional fixed-fee models towards performance-based agreements. Payment is linked to inspection items delivered and measurable value achieved:

- No results, no payment.
- Every inspection delivers measurable impact.
- Accountability is built into the model.

### **Looking Ahead: A Future Shaped by People, Data, and Purpose**

The enriched RBI strategy represents more than a scheduling change—it is a transformation of mindset, method, and mission. Inspections are redefined not as a regulatory requirement but as a strategic enabler of operational excellence, safety assurance, and long-term resilience.

Through investment in people, the development of bespoke risk algorithms, and the integration of real-time digital intelligence, RBI provides sharper decision-making, more efficient workflows, and stronger asset performance across the offshore lifecycle.

This forms a new foundation for Ex compliance:

**Data-led. Risk-informed. People-first.**

### **The Shift: From Compliance to Intelligence**

Time-based inspection regimes have been replaced by context-driven, risk-informed logic. Environmental conditions, criticality, and historical performance all feed into the prioritisation process. Every inspection result enhances the risk engine, creating a living, adaptive surveillance layer.

Crucially, technology is only one part of the equation. Engineers, inspectors, analysts, and decision-makers are central to converting insight into action and data into safety.

## Smart Services: A Future-Proof Digital Offering

Through its dedicated Smart Services line, the organisation provides offshore operators with a scalable, intelligent support structure that unites deep domain expertise with advanced digital integration:

- **Connect** — Secure, instant access to documentation, spare parts information, risk insights, and asset history.
- **Predict** — AI-driven agents that identify anomalies, anticipate failure trends, and enable proactive maintenance planning.
- **Elevate** — Visual dashboards, risk modelling, and expert-informed recommendations that guide strategic action.

These services are not static; they evolve in line with asset requirements, user behaviour, and emerging operational threats. With every inspection cycle, the RBI platform increases in intelligence, transforming it from a digital tool into a trusted partner in operational success.

## Conclusion: Smarter Inspections. Safer Assets. Stronger Outcomes

Risk-Based Inspection is not simply about reducing effort—it is about achieving stronger, safer, and more efficient outcomes. It equips frontline technicians with the tools they need, empowers engineering teams with actionable insight, and provides leadership with the evidence required to act decisively.

In offshore environments—where operating conditions are harsh and safety is paramount—RBI represents a forward-looking strategy. It is built upon intelligent data, human expertise, and a commitment to continuous improvement.

The industry is entering a new era in which:

- Risk is no longer assumed—it is measured, modelled, and actively managed in real time.
- Data is no longer hidden in static spreadsheets—it is live, visual, and ready to inform decisions.
- Compliance is no longer reactive—it is proactive, dynamic, and value-generating.

## A People-First Future: Built Together

The transition from manual, reactive processes to digital, risk-based inspection is reshaping hazardous area management. Yet it is people who remain central to making this transformation real.

By equipping individuals with sharper data, more intelligent tools, and a culture of collaboration, the organisation is creating a future where:

- Every inspection delivers value.
- Every decision is evidence-based.
- Every preventable failure is avoided.
- Every action creates lasting impact.

Innovation in Ex inspection is not about replacing people—it is about enabling them to perform at their best. With the right digital systems and risk insights, teams are able to make faster decisions, act more safely, and deliver stronger operational results.

This represents the future of Ex inspection: a partnership between people, data, and purpose—working in harmony to protect life, minimise risk, and strengthen industry sustainability.

## What Comes Next?

The organisation remains committed to advancing its approach—developing predictive asset lifecycle models, refining AI-driven risk tools, and enabling campaign strategies that deliver measurable improvements.

It will continue to partner with operators—sharing knowledge, co-developing intelligent solutions, and supporting them at every stage of their RBI journey.

Above all, it continues to uphold the belief that when people, data, and technology are combined with clear intent, assets are not merely inspected—

They are **protected, optimised, and elevated.**

**The future of Ex inspection is here. Together, it can be built.**